



MANIPAL

ACADEMY *of* HIGHER EDUCATION

(Institution of Eminence Deemed to be University)

MANIPAL SCHOOL OF INFORMATION SCIENCES

(A Constituent unit of MAHE, Manipal)

CYE 5101 – 2_Theory_HW

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MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

1. Encrypt “CHASE YOUR DREAMS”

a. Caesar Cipher

$$C = (P+3) \bmod 26$$

Plaintext	(P+3) mod 26	Ciphertext
C	5	F
H	10	K
A	3	D
S	21	V
E	7	H
Y	1	B
O	17	R
U	23	X
R	20	U
D	6	G
R	20	U
E	7	H
A	3	D
M	15	P
S	21	V

Ciphertext: **FKDVH BRXU GUHDPV**

b. Shift Cipher (7)

$$C = (P+7) \bmod 26$$

Plaintext	(P+7) mod 26	Ciphertext
C	9	J
H	14	O
A	7	H
S	25	Z
E	11	L
Y	5	F
O	21	V
U	1	B
R	24	Y
D	10	K
R	24	Y

E	11	L
A	7	H
M	19	T
S	25	Z

Ciphertext: **JOHZL FVBY KYLHTZ**

c. ROT13

$$C = (P+13) \bmod 26$$

Plaintext	$(P+13) \bmod 26$	Ciphertext
C	15	P
H	20	U
A	13	N
S	5	F
E	17	R
Y	11	L
O	1	B
U	7	H
R	4	E
D	16	Q
R	4	E
E	17	R
A	13	N
M	25	Z
S	5	F

Ciphertext: **PUNFR LBHE QERNZF**

2. Decrypt:

a. **WRWDOOB VHFUHW PHVVDJH** using Caesar Cipher

$$P = (C-3) \bmod 26$$

Ciphertext	$(C-3) \bmod 26$	Plaintext
W	19	T
R	14	O
W	19	T
D	0	A
O	11	L

O	11	L
B	24	Y
V	18	S
H	4	E
F	2	C
U	17	R
H	4	E
W	19	T
P	12	M
H	4	E
V	18	S
V	18	S
D	0	A
J	6	G
H	4	E

Plaintext: **TOTALLY SECRET MESSAGE**

- b. 1) **MJQQT BTWQI** using Shift Cipher 5

$$P = (C-5) \bmod 26$$

Ciphertext	(C-5) mod 26	Plaintext
M	7	H
J	4	E
Q	11	L
Q	11	L
T	14	O
B	22	W
T	14	O
W	17	R
Q	11	L
I	3	D

Plaintext: **HELLO WORLD**

- 2) **SJB DTWP YNRJ** using Shift Cipher 5

$$P = (C-5) \bmod 26$$

Ciphertext	(C-5) mod 26	Plaintext
S	13	N
J	4	E
B	22	W
D	24	Y
T	14	O
W	17	R
P	10	K
Y	19	T
N	8	I
R	12	M
J	4	E

Plaintext: **NEW YORK TIME**

- c. **FRAQ ZBER PBSSRR** using ROT13

$$P = (C-13) \bmod 26$$

Ciphertext	(C-13) mod 26	Plaintext
F	18	S
R	4	E
A	13	N
Q	3	D
Z	12	M
B	14	O
E	17	R
R	4	E
P	2	C
B	14	O
S	5	F
S	5	F
R	4	E
R	4	E

Plaintext: **SEND MORE COFFEE**